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Inventor(s)

Wei; Yong-Chao

Method for manufacturing a circuit board with one embedded electronic component

Abstract

A method for manufacturing a circuit board is provided, which includes a wiring board. The wiring board includes a first wiring layer, a dielectric layer and a second wiring layer stacked, and conductive pillars. Each conductive pillar connects the first wiring layer and the second wiring layer. A groove is recessed from a side of the dielectric layer facing away from the second wiring layer, and includes first recessed portion and second recessed portions recessed from a sidewall of the first recessed portion. An end surface of each conductive pillar is exposed from the at least two spaced second recessed portions, and a sidewall of each pillar close to the first recessed portion is exposed from the second recessed portion. An one electronic component is received in the first recessed portion, and is connected to the conductive pillars through electrical connecting portions received in the second recessed portions.

Inventors: Wei; Yong-Chao (Qinhuangdao, CN)

Applicant: QING DING PRECISION ELECTRONICS (HUAIAN) CO., LTD (Huai an, CN); Avary Holding (Shenzhen) Co., Limited. (Shenzhen, CN); GARUDA TECHNOLOGY CO., LTD. (New Taipei, TW)

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Assignee: QING DING PRECISION ELECTRONICS (HUAIAN) CO., LTD (Huai an, CN); Avary Holding (Shenzhen) Co., Limited. (Shenzhen, CN); GARUDA TECHNOLOGY CO., LTD. (New Taipei, TW)

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Primary Examiner: Trinh; Minh N

Attorney, Agent or Firm: ScienBiziP, P.C.

Background/Summary

FIELD

(1) The subject matter herein generally relates to a circuit board, especially relates to a circuit board with at least one embedded electronic component and a method for manufacturing the circuit board with the embedded electronic component.

BACKGROUND

(2) Existing embedded circuit boards usually use SMT (Surface Mount Technology) solder paste to solder electronic components on the surface of a substrate. Since the solder paste occupies a certain thickness, it is not conducive to the thinning of the circuit board. However, other existing processes for fixing the electronic component inside the circuit board from the side surface are often more complicated.

(3) Therefore, there is room for improvement within the art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Implementations of the present disclosure will now be described, by way of embodiments, with reference to the attached figures.

(2) FIG. 1 is a flowchart of an embodiment of a method for manufacturing a circuit board.

(3) FIG. 2 is a cross-sectional view of an embodiment of a wiring board.

(4) FIG. 3A is a cross-sectional view showing a mask with at least one first opening and at least two spaced second openings on the wiring board of FIG. 2.

(5) FIG. 3B is a top view of the mask on the wiring board of FIG. 3A.

(6) FIG. 4A is a cross-sectional view showing a groove on the wiring board of FIG. 3A.

(7) FIG. 4B is a top view of the wiring board with the groove of FIG. 4A.

(8) FIG. 5 is a cross-sectional view showing an electronic component in the groove of FIG. 4A.

(9) FIG. 6A is a cross-sectional view showing electrical connecting portions electrically connect the electronic component and the wiring board of FIG. 5.

(10) FIG. 6B is a top view of the wiring board with the electrical connecting portions of FIG. 6A.

(11) FIG. 7 is a cross-sectional view showing the mask peeled off from the wiring board of FIG. 6A.

(12) FIG. 8 is a cross-sectional view showing outer wiring structures on the wiring board of FIG. 7.

(13) FIG. 9 is a cross-sectional view showing solder masks on the wiring board of FIG. 8.

(14) FIG. 10 is a flowchart of an embodiment of a method for manufacturing a wiring board.

(15) FIG. 11 is a cross-sectional view of an embodiment of a double-sided copper clad laminate including a first copper foil, a first insulating layer and a second copper foil.

(16) FIG. 12 is a cross-sectional view showing a third wiring layer a plurality of spaced first conductive portions on the first insulating layer of FIG. 11.

(17) FIG. 13 is a cross-sectional view showing a first single-side copper clad laminate including a second insulating layer and a third copper foil on the first insulating layer of FIG. 12.

(18) FIG. 14 is a cross-sectional view showing a second wiring layer on the second insulating layer of FIG. 13.

(19) FIG. 15 is a cross-sectional view showing a second single-side copper clad laminate including a third insulating layer and a fourth copper foil on the first insulating layer of FIG. 14.

(20) FIG. 16 is a cross-sectional view showing connecting holes on the second single-side copper clad laminate of FIG. 15.

(21) FIG. 17 is a cross-sectional view showing two electronic components in the groove of FIG. 4A.

(22) FIG. 18 a top view of the wiring board with the two electronic components of FIG. 17.

(23) FIG. 19 is a cross-sectional view of an embodiment of a circuit board.

(24) FIG. 20 is a cross-sectional view of another embodiment of a circuit board.

DETAILED DESCRIPTION

(25) It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous

elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, methods, procedures, and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale, and the proportions of certain parts may be exaggerated to better illustrate details and features of the present disclosure.

(26) The term “comprising,” when utilized, means “including, but not necessarily limited to”; it specifically indicates open-ended inclusion or membership in the so-described combination, group, series, and the like.

(27) FIG. 1 illustrates a flowchart of a method in accordance with an embodiment. The embodiment method for manufacturing a circuit board with at least one embedded electronic component is provided by way of embodiments, as there are a variety of ways to carry out the method. Each block shown in FIG. 1 represents one or more processes, methods, or subroutines carried out in the method. Furthermore, the illustrated order of blocks can be changed. Additional blocks may be added or fewer blocks may be utilized, without departing from this disclosure. The method can begin at block 201.

(28) At block 201, referring to FIG. 2, a wiring board 10 is provided. The wiring board 10 includes a dielectric layer 11, a first wiring layer 13, a second wiring layer 14, and a plurality of spaced conductive pillars 15. The first wiring layer 13 and the second wiring layer 14 are formed on opposite sides of the dielectric layer 11. Each of the plurality of spaced conductive pillars 15 penetrates the dielectric layer 11 and electrically connects the first wiring layer 13 and the second wiring layer 14.

(29) The wiring board 10 may be a double-layer wiring board or a multilayer wiring board.

(30) In at least one embodiment, the wiring board 10 is multilayer wiring board. Specifically, the wiring board 10 further includes a third wiring layer 12a embedded in the dielectric layer 11 and located between the first wiring layer 13 and the second wiring layer 14. Each of the plurality of spaced conductive pillars 15 may include a second conductive portion 153, a first conductive portion 151, and a third conductive portion 156 connected in that sequence. The second conductive portion 153 and the first conductive portion 151 are connected between the first wiring layer 13 and the third wiring layer 12a. The third conductive portion 156 is connected between the second wiring layer 14 and an end of the first conductive portion 151 facing away from the first wiring layer 13.

(31) At block 202, referring to FIGS. 3A and 3B, a mask 30 is attached to a side of the wiring board 10, and at least one first opening 31 exposing a part of the dielectric layer 11 and at least two spaced second openings 33 are formed on the mask 30. Each second opening 33 is recessed from a sidewall of the first opening 31 toward a direction away from a center axis of the first opening 31, and arranged corresponding to one of the plurality of spaced conductive pillars 15 to expose at least a part of an end surface of the corresponding conductive pillar 15.

(32) In at least one embodiment, the mask 30 is attached to a side of the wiring board 10 facing away from the second wiring layer 14.

(33) In at least one embodiment, a shape of each first opening 31 and a shape of each second opening 33 are both rectangular. A distance that the second opening 33 is recessed from the sidewall of the first opening 31 is defined as a width of the second opening 33, and a size of the second opening 33 in a direction perpendicular to the width is defined as a length of the second opening 33. In another embodiment, the shape of each first opening 31 and the shape of each second opening 33 may be both varied as needed, for example, may be regular shapes such as ellipse, circle, sector, polygon, or may be irregular shapes.

(34) At block 203, referring to FIGS. 4A and 4B, a part of the dielectric layer 11 exposed from each

of the at least one first opening **31** and the second openings **33** communicating with the first opening **31** is removed to form a groove **40**, and the groove **40** does not penetrate the dielectric layer **11**. Each groove **40** includes a first recessed portion **41** and at least two spaced second recessed portions **43**. The first recessed portion **41** corresponds to the first opening **31**. Each second recessed portion **43** is recessed from a sidewall of the first recessed portion **41** toward a direction away from a center axis of the first recessed portion **41**. Each second recessed portion **43** corresponds to one of the second openings **33** to expose a part of a side wall of the corresponding conductive pillar **15** close to the first recessed portion **41**.

(35) In at least one embodiment, a shape of each first recessed portion **41** and a shape of each second recessed portion **43** are both rectangular. In another embodiment, the shape of each first recessed portion **41** and the shape of each second recessed portion **43** may be both varied as needed, for example, may be regular shapes such as ellipse, circle, sector, polygon, or may be irregular shapes.

(36) At block **204**, referring to FIG. **5**, at least one electronic component **50** is placed in the at least one first recessed portion **41**. Each of the at least one electronic component **50** includes at least two spaced connecting pads **51**. Each of the at least two spaced connecting pads **51** corresponds to one of the plurality of spaced conductive pillars **15** in the groove **40** exposing the connecting pads **51**.

(37) At block **205**, referring to FIGS. **6A** and **6B**, each of the at least two spaced second recessed portions **43** of each groove **40** is filled with a conductive material to form an electrical connecting portion **55** to electrically connect one of the plurality of spaced conductive pillars **15** corresponding to the recessed portion **43** and the corresponding connecting pad **51**.

(38) The conductive material may be soldering flux such as tin paste, which is melted and solidified to form the electrical connecting portion **55** to connect the corresponding conductive pillar **15** and the corresponding connecting pad **51**. The conductive material may be conductive glue. The conductive glue fills in the second recessed portion **43** and is cured to form the electrical connecting portion **55**.

(39) At block **206**, referring to FIG. **7**, the mask **30** is peeled off from the wiring board **10** with the electrical connecting portions **55** and the at least one electronic component **50**.

(40) In at least one embodiment, the method for manufacturing a circuit board with at least one embedded electronic component may further include the following blocks **207** and **208**.

(41) At block **207**, referring to FIG. **8**, an outer wiring structure **60** is formed on the side of the wiring board **10** where the groove **40** is provided to encapsulate the at least one electronic component **50** in the at least one groove **40**.

(42) In at least one embodiment, two outer wiring structures **60** are formed on two opposite sides of the wiring board **10**. Each of the outer wiring structures **60** is single-layer wiring board. In another embodiment, each of the outer wiring structures **60** may be double-layer wiring board or multilayer wiring board.

(43) When forming the outer wiring structure **60**, gaps between the at least one groove **40** and the at least one electronic component **50** is filled with dielectric materials of the dielectric layer **11** and the outer wiring structure **60** during a pressing process.

(44) At block **208**, referring to FIG. **9**, two solder masks **70** are respectively formed on the opposite sides of the wiring board **10**, a side of the outer wiring structure **60** facing away from the wiring board **10** is covered by one of the solder masks **70**.

(45) In at least one embodiment, each of the outer wiring structures **60** on the opposite sides of the wiring board **10** is covered by one of the solder masks **70**.

(46) FIG. **10** illustrates a flowchart of an embodiment of a method for manufacturing the wiring board **10** (shown in FIG. **1**). The method can begin at block **801**.

(47) At block **801**, referring to FIG. **11**, a double-sided copper clad laminate **10a** is provided. The double-sided copper clad laminate **10a** includes a first copper foil **121**, a first insulating layer **11a** and a second copper foil **122** stacked in that sequence along a first direction.

(48) In at least one embodiment, the first insulating layer **11a** is made of a developing material, such as a developable photoresist or a developing ink. In another embodiment, the first insulating layer **11a** may be made of other dielectric materials commonly used in the art, such as phenolic resin, epoxy resin (EP), polyimide resin (PI), polyester resin (PET), polyphenylene oxide resin (PPO), polytetrafluoroethylene resin (PTFE), or bismaleimide triazine resin (BT).

(49) At block **802**, referring to FIG. **12**, a third wiring layer **12a** is formed by performing a circuit fabrication process on the first copper foil **121**, and a plurality of spaced first conductive portions **151** is formed. Each of the plurality of spaced first conductive portions **151** penetrates the first insulating layer **11a** and connects the second wiring layer **12a** and the second copper foil **122**.

(50) In at least one embodiment, a width of a cross section along the first direction of each of the plurality of spaced first conductive portions **151** may gradually decrease from the third wiring layer **12a** to the second copper foil **122**. Therefore, it is convenient to subsequently fill conductive materials to form the electrical connecting portions **55**. In at least one embodiment, the cross-section along the first direction of each of the plurality of spaced first conductive portions **151** may be trapezoidal. In at least one embodiment, each of the plurality of spaced first conductive portions **151** may be a circular truncated cone.

(51) At block **803**, referring to FIG. **13**, a first single-side copper clad laminate **10b** is pressed on a side of the first insulating layer **11a** facing away from the second copper foil **122**. The first single-side copper clad laminate **10b** includes a second insulating layer **11b** combined with the third wiring layer **12a** and a third copper foil **123** formed on the second insulating layer **11b** facing away from the third wiring layer **12a**.

(52) In at least one embodiment, the second insulating layer **11b** may be made of phenolic resin, epoxy resin, polyimide resin, polyester resin, polyphenylene oxide resin, polytetrafluoroethylene resin, or bismaleimide triazine resin.

(53) At block **804**, referring to FIG. **14**, a second wiring layer **14** is formed by performing a circuit fabrication process on the third copper foil **123**, and the second copper foil **122** is removed.

(54) In at least one embodiment, a third conductive portion **156** is formed to connect an end of one of the plurality of spaced first conductive portions **151** facing the second wiring layer **14** and the second wiring layer **14**.

(55) At block **805**, referring to FIG. **15**, a second single-side copper clad laminate **10c** is pressed on a side of the first insulating layer **11a** facing away from the second wiring layer **14**. The second single-side copper clad laminate **10c** includes a third insulating layer **11c** combined with the first insulating layer **11a** and a fourth copper foil **124** formed on a side of the third insulating **11c** facing away from the first insulating layer **11a**.

(56) In at least one embodiment, the third insulating layer **11c** may be made of a developing material, such as a developable photoresist or a developing ink. In another embodiment, the third insulating layer **11c** may be made of other dielectric materials commonly used in the art, such as phenolic resin, epoxy resin (EP), polyimide resin (PI), polyester resin (PET), polyphenylene oxide resin (PPO), polytetrafluoroethylene resin (PTFE), or bismaleimide triazine resin (BT).

(57) Preferably, the third insulating layer **11c** and the first insulating layer **11a** are both made of the same materials.

(58) At block **806**, referring to FIG. **16**, a connecting hole **101c** corresponding to each of the plurality of spaced first conductive portions **151** is formed on the second single-side copper clad laminate **10c** to expose an end of each of the plurality of spaced first conductive portions **151** facing away from the second wiring layer **14**.

(59) In at least one embodiment, in a cross section along the first direction, a maximum width R1 of each connecting hole **101c** is less than or equal to a width R2 of the end of the corresponding conductive portion **151** facing away from the second wiring layer **14**. In at least one embodiment, R1 is equal to R2, and each connecting hole **101c** is cylindrical. In another embodiment, the shape of each connecting hole **101c** may be varied as needed.

- (60) The connecting hole **101c** may be formed by laser cutting or mechanical drilling.
- (61) At block **807**, referring to FIG. **2**, a first wiring layer **13** is formed by performing a circuit fabrication process on the fourth copper foil **124**, and a second conductive portion **153** corresponding to each connecting hole **101c** is formed to fill the connecting hole **101c** and connect the corresponding first conductive portion **151**, thereby obtaining the wiring board **10**. Each first conductive portion **151** connects the corresponding second conductive portion **153** and the corresponding third conductive portion **156** to form a conductive pillar **15**. The first insulating layer **11a**, the second insulating layer **11b**, and the third insulating layer **11c** are formed a dielectric layer **11**.
- (62) A maximum width **R1** of the second conductive portion **153** is less than or equal to the width **R2** of the end of the corresponding first conductive portion **151** facing away from the second wiring layer **14**.
- (63) When the first insulating layer **11a** and the third insulating layer **11c** are both made of developing materials, the part of the dielectric layer **11** exposed from each of the at least one first opening **31** and the second openings **33** is removed by exposure and development to form the groove **40**. In at least one embodiment, at least a part of each first conductive portion **151** corresponding to the second openings **33** is exposed from the corresponding second recessed portion **43**.
- (64) An area of the third insulating layer **11c** exposed from the at least one first opening **31** and the at least two spaced second openings **33** and an area of the first insulating layer **11a** corresponding to the at least one first opening **31** and the at least two spaced second openings **33**, are removed by exposure and development. A thickness of the first insulating layer **11a** and a thickness of the third insulating layer **11c** may be adjusted as needed, so as to facilitate a subsequent adjustment of a depth of the groove **40** to accommodate the embedding of electronic components of different thicknesses.
- (65) A width of an end of the conductive portion **151** facing the second wiring layer **14** is defined as **R3**. Preferably, the length of the second opening **33** is greater than or equal to **R3**. The width of the second opening **33** is less than or equal to $R3 + (R3 - R2)/2$, and also greater than or equal to $(R3 - R2)/2$. In at least one embodiment, the length of the second opening **33** is **R3**, and width of the second opening **33** is $R3 + (R3 - R2)/2$.
- (66) In at least one embodiment, referring to FIGS. **6A** and **6B**, each of the at least one groove **40** receives one electronic component **50**. In another embodiment, referring to FIGS. **17** to **19**, each of the at least one groove **40** receives at least two electronic components **50**. Referring to FIGS. **6B** and **18**, when each of the at least one groove **40** corresponds to a plurality of conductive pillars **15** and receives at least two electronic components **50**, an arrangement of the at least two electronic components **50** may be adjusted based on an arrangement of the plurality of conductive pillars **15** and the actual need.
- (67) Depending on the embodiment, certain of the steps of methods described may be removed, others may be added, and the sequence of steps may be altered. It is also to be understood that the description and the claims drawn to a method may include some indication in reference to certain steps. However, the indication used is only to be viewed for identification purposes and not as a suggestion as to an order for the steps.
- (68) FIG. **20** illustrates an embodiment of a circuit board **100a** with at least one embedded electronic component. The circuit board **100** includes a wiring board **10** and at least one electronic component **50**. The wiring board **10** includes a dielectric layer **11**, a plurality of spaced conductive pillars **15**, a first wiring layer **13**, and a second wiring layer **14**. The first wiring layer **13** and the second wiring layer **14** are stacked along a first direction, and formed on opposite sides of the dielectric layer **11**. Each of the plurality of spaced conductive pillars **15** electrically connects the first wiring layer **13** and the second wiring layer **14**.
- (69) At least one groove **40** is recessed from a side of the dielectric layer **11** facing away from the

second wiring layer **14** toward the second wiring layer **14**. Each of the at least one groove **40** includes a first recessed portion **41** and at least two spaced second recessed portions **43**. Each of the at least two spaced second recessed portions **43** is recessed from a sidewall of the first recessed portion **41** toward a direction away from the first recessed portion **41**. At least a part of an end surface of each of the plurality of spaced conductive pillars **15** facing away from the second wiring layer **14** is exposed from the at least two spaced second recessed portions **43**, a part of a sidewall of each of the plurality of spaced conductive pillars **15** close to the first recessed portion **41** is exposed from the at least two spaced second recessed portions **43**. The at least one electronic component **50** is received in the first recessed portion **41**, and is electrically connected to the plurality of spaced conductive pillars **15** through electrical connecting portions **55** received in the at least two spaced second recessed portions **43**.

(70) Each of the plurality of spaced conductive pillars **15** includes a first conductive portion **151**. A width of a cross section along the first direction of the conductive portion **151** may gradually decrease from an end of the conductive portion **151** facing the wiring layer **14** to an end of the conductive portion **151** facing away from the wiring layer **14**. Therefore, it is convenient to form the electrical connecting portions **55**.

(71) The dielectric layer **11** includes a first layer **111** and a second layer **112** stacked along the first direction. The first layer **111** may be made of a developing material, such as a developable photoresist or a developing ink.

(72) In at least one embodiment, each of the at least one groove **40** penetrates the first layer **111**.

(73) The wiring board **10** may be a double-layer wiring board or a multilayer wiring board. In at least one embodiment, the wiring board **10** is a three-layer wiring board.

(74) In at least one embodiment, each of the at least one groove **40** receives one electronic component **50**. In another embodiment, each of the at least one groove **40** receives at least two electronic components **50**, and an arrangement of the at least two electronic components **50** may be varied as needed.

(75) In the above method of for manufacturing a circuit board, the electronic component is embedded into an area of the circuit board around the plurality of the conductive pillars and electrically connects the plurality of the conductive pillars. Compared with the prior art, the above method may omit a process of forming conductive areas on sidewalls to connect the electronic component and the wiring layer. The number and distribution of the embedded electronic components may be adjusted according to the circuit design, which is beneficial to improve a flexibility of embedding the electronic components. In addition, the above method of for manufacturing the circuit board is simple in process and easy to produce, and is beneficial to the lightness and thinness of the embedded circuit board.

(76) It is to be understood, even though information and advantages of the present embodiments have been set forth in the foregoing description, together with details of the structures and functions of the present embodiments, the disclosure is illustrative only; changes may be made in detail, especially in matters of shape, size, and arrangement of parts within the principles of the present embodiments to the full extent indicated by the plain meaning of the terms in which the appended claims are expressed.

Claims

1. A method for manufacturing a circuit board comprising: providing a wiring board comprising a dielectric layer, a first wiring layer, a second wiring layer, and a plurality of spaced conductive pillars, the first wiring layer and the second wiring layer formed on opposite sides of the dielectric layer, each of the plurality of spaced conductive pillars penetrating the dielectric layer and electrically connecting the first wiring layer and the second wiring layer; attaching a mask on a side of the wiring board, and forming a first opening exposing a part of the dielectric layer and at least

two spaced second openings on the mask, wherein each of the at least two spaced second openings is recessed from a sidewall of the first opening toward a direction away from the first opening, and is arranged corresponding to one of the plurality of spaced conductive pillars to expose at least a part of an end surface of the corresponding conductive pillar; removing a part of the dielectric layer exposed from the first opening and the at least two spaced second openings to form a groove, and the groove not penetrating the dielectric layer, wherein the groove comprises a first recessed portion and at least two spaced second recessed portions, the first recessed portion corresponds to the first opening, each of the at least two spaced second recessed portions is recessed from a sidewall of the first recessed portion toward a direction away from first recessed portion, each of the at least two spaced second recessed portions corresponds to one of the second openings to expose a part of a side wall of the corresponding conductive pillar close to the first recessed portion; placing a single electronic component in the first recessed portion, wherein the single electronic component comprises at least two spaced connecting pads, each of the at least two spaced connecting pads corresponds to one of the plurality of spaced conductive pillars in the groove; and filling a conductive material in each of the at least two spaced second recessed portions to form an electrical connecting portion to electrically connect one of the at least two spaced connecting pads and the corresponding conductive pillar.

2. The method of claim 1, wherein the providing of the wiring board further comprises steps of: providing a double-sided copper clad laminate comprising a first copper foil, a first insulating layer and a second copper foil stacked in that sequence along a first direction; forming a third wiring layer by performing a circuit fabrication process on the first copper foil, and forming a plurality of spaced first conductive portions, wherein each of the plurality of spaced first conductive portions penetrates the first insulating layer and connects the second wiring layer and the second copper foil; pressing a first single-side copper clad laminate on a side of the first insulating layer facing away from the second copper foil, the first single-side copper clad laminate comprising a second insulating layer combined with the third wiring layer and a third copper foil formed on the second insulating layer facing away from the third wiring layer; forming a second wiring layer by performing a circuit fabrication process on the third copper foil, forming a third conductive portion between the second wiring layer and the third wiring layer to connect the second wiring layer and the one of the plurality of spaced first conductive portions, and removing the second copper foil; pressing a second single-side copper clad laminate on a side of the first insulating layer facing away from the second wiring layer, the second single-side copper clad laminate comprising a third insulating layer combined with the first insulating layer and a fourth copper foil formed on a side of the third insulating facing away from the first insulating layer; forming a connecting hole corresponding to each of the plurality of spaced first conductive portions on the second single-side copper clad laminate to expose an end of each of the plurality of spaced first conductive portions facing away from the second wiring layer; and forming a first wiring layer by performing a circuit fabrication process on the fourth copper foil, and forming a second conductive portion corresponding to each connecting hole to fill the connecting hole and connect the corresponding first conductive portion, thereby obtaining the wiring board, wherein each second conductive portion connects the corresponding second conductive portion and the corresponding third conductive portion to form a conductive pillar, the first insulating layer, the second insulating layer, and the third insulating layer are formed a dielectric layer.

3. The method of claim 2, wherein a material of the first insulating layer and a material of the third insulating layer are chosen from a same material comprising developing materials.

4. The method of claim 3, wherein a part of the dielectric layer exposed from the first opening and the at least two spaced second openings to form a groove further comprising: removing an area of the third insulating layer exposed from the first opening and the at least two spaced second openings and an area of the first insulating layer corresponding to first opening and the at least two spaced second openings by exposure and development.

5. The method of claim 2, wherein a width of a cross section along the first direction of each of the plurality of spaced first conductive portions gradually decreases from the third wiring layer to the second copper foil, a width of a cross section along the first direction of each second conductive portion is less than or equal to a minimum width of the cross section along the first direction of the first conductive portion connecting the second conductive portion.
 6. The method of claim 1, further comprising: forming an outer circuit structure on the wiring board to encapsulate the single electronic component in the groove.
 7. The method of claim 6, further comprising: forming two solder masks on opposite sides of the wiring board, wherein a side of the outer circuit structure facing away from the wiring board is combined with one of the solder masks.
 8. The method of claim 1, wherein the attaching of the mask on a side of the wiring board further comprises attaching the mask to a side of the wiring board facing away from the second wiring layer.
 9. The method of claim 1, wherein the conductive material is a conductive glue.
 10. The method of claim 1, wherein a maximum width of the second conductive portion is less than or equal to a width of an end of a corresponding one of the plurality of spaced first conductive portions facing away from the second wiring layer.
 11. The method of claim 1, wherein the first recessed portion and each of the at least two spaced second recessed portions are rectangular shapes.
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